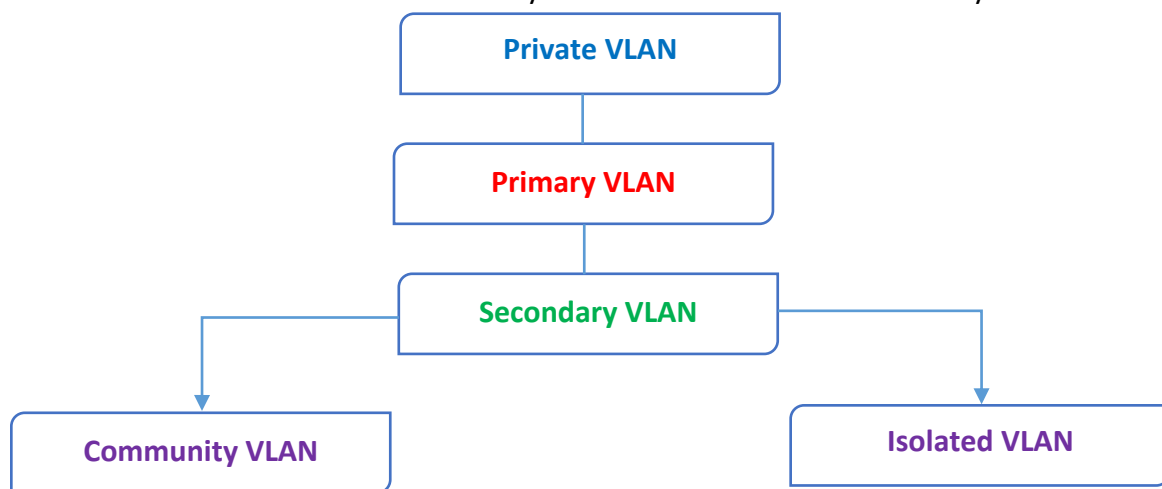


Private VLAN:

- o In Cisco World PVLAN stands for Private Virtual Local Area Network.
- o Private VLAN (PVLAN) offers another level of security in Cisco Switches.
- o Private VLAN (PVLAN) is used to further divide a single VLAN into Sub-VLANs.
- o Private VLAN that enables an organization to further secure the VLAN ports.
- o Private VLAN are used to break Layer 2 broadcast domain into small subdomains.
- o PVLANS require Layer 3 device to communicate with some other VLAN.
- o PVLANS provide Layer 2 isolation between ports within the same broadcast domain.
- o Private VLANs feature can be used to create Secondary VLANs inside a Primary VLAN.
- o Private VLAN are used to break the Layer 2 broadcast domain into small subdomains.
- o The Subdomain consists of one Primary VLAN and one or more Secondary VLAN.



Types of VLAN:

There are two types of VLANs in Private VLANs, **Primary VLAN** and **Secondary VLAN**.

Primary VLAN:

- o Primary VLAN is just normal and regular Virtual Local Area Network (VLAN).
- o Private VLANs feature can be used to create Secondary VLANs inside Primary VLAN.
- o The Private VLAN always has one Primary Virtual Local Area Network (VLAN).
- o In Private VLAN, Primary Virtual Local Area Network (VLAN) have Promiscuous Port.
- o Primary VLAN have one or more Secondary Virtual Local Area Network (VLAN).
- o Primary VLAN is used for receiving traffic from device connected to Promiscuous Port.
- o All the ports in the Private VLAN belongs to a Primary Virtual Local Area Network VLAN.
- o All VLANs in Private VLAN domain shares same Primary Virtual Local Area Network VLAN.

Secondary VLAN:

- o A Private VLAN can have one or more Secondary Virtual Local Area Network VLANs.
- o Secondary VLAN provides isolation between the ports belonging to same Private VLAN.
- o It provides isolation between the ports belonging to same private VLAN domain.

Type of Secondary VLAN:

Isolated VLAN:

- o Ports associated with an isolated private VLAN **cannot** communicate with one another.
- o They can communicate with a Promiscuous Port within the same Private VLAN (PVLAN).

Community VLAN:

- o Ports associated with Community VLAN **can** communicate with one another.
- o They can also communicate with a Promiscuous port within the Private VLAN (PVLAN).
- o Community VLAN forwards traffic within community ports & to the promiscuous ports.

PVLAN Port Types:

- o There are three types of PVLAN ports Promiscuous, Isolated, and Community.

Promiscuous Port:

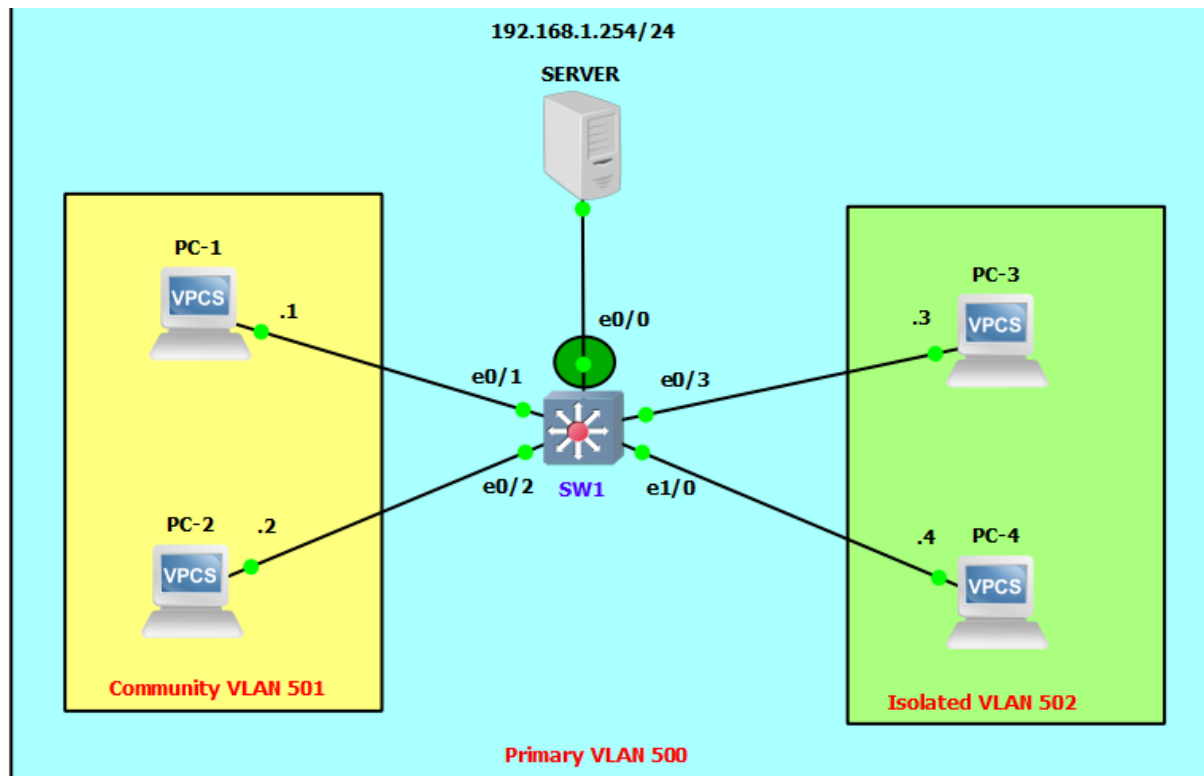
- o Promiscuous Port is usually connects to a router or connected to default Devices.
- o Promiscuous Port is allowed to send & receive frames from any other port on VLAN.
- o A Promiscuous Port can communicate with all interfaces inside the Private VLAN.

Isolated Port:

- o Isolated Port can only allowed communicating with Promiscuous ports.
- o Isolated Port cannot communicate with other ports within the same PVLAN.
- o PVLANS block all traffic to isolated ports except traffic from promiscuous ports.
- o Isolated Ports as the name suggests isolated and usually connects to hosts.

Community Port:

- o Community Port is used for group of users who want communication between them.
- o Community ports are allowed to talk to their Community ports.
- o Community ports talk with themselves and only with Promiscuous ports.
- o Community ports cannot communicate with other communities or isolated ports.



Basic IP Configuration	
PC-1> ip 192.168.1.1 255.255.255.0	PC-3> ip 192.168.1.3 255.255.255.0
PC-2> ip 192.168.1.2 255.255.255.0	PC-4> ip 192.168.1.4 255.255.255.0
SERVER> ip 192.168.1.254 255.255.255.0	SW1(config)#interface vlan 1
	SW1(config-if)#192.168.1.100 255.255.255.0

Private VLAN Configuration
SW1(config)#vtp mode transparent
SW1(config)#vlan 501 SW1(config-vlan)#private-vlan community SW1(config-vlan)#exit
SW1(config)#vlan 502 SW1(config-vlan)#private-vlan isolated SW1(config-vlan)#exit
SW1(config-vlan)#vlan 500 SW1(config-vlan)#private-vlan primary SW1(config-vlan)#private-vlan association 501,502
SW1(config)#interface ethernet 0/0 SW1(config-if)#switchport mode private-vlan promiscuous SW1(config-if)#switchport private-vlan mapping 500 501,502
SW1(config)#interface rang ethernet 0/1-2 SW1(config-if-range)#switchport mode private-vlan host SW1(config-if-range)#switchport private-vlan host-association 500 501
SW1(config)#int rang e0/3,e1/0 SW1(config-if-range)#switchport mode private-vlan host SW1(config-if-range)#switchport private-vlan host-association 500 502
SW1#show vlan private-vlan
SW1#show vlan private-vlan type
SW1#show interfaces e0/0 switchport
SW1#show interfaces e0/1 switchport
SW1#show interfaces e0/2 switchport

SW1#show vlan private-vlan

Primary	Secondary	Type	Ports
500 500	501 502	community isolated	Et0/0, Et0/1, Et0/2 Et0/0, Et0/3, Et1/0